

On Weak Generation

Theorem 1. *Let*

$$\mathcal{C} = D(\text{Comod}_{\mathcal{O}(\mathbb{G}_a)}), \quad \widehat{\mathcal{C}} = \varprojlim_{q \geq 0} \mathcal{C}^{\geq -q},$$

where $k = \mathbb{F}_p$ is the trivial comodule and $\widehat{\mathcal{C}}$ is the left completion of the standard t -structure. Then there is a nonzero $X \in \widehat{\mathcal{C}}$ such that

$$\text{RHom}_{\widehat{\mathcal{C}}}(k, X) \simeq 0.$$

Proof. Put $R = \text{End}_{\mathcal{C}}(k) \in \text{CAlg}(\text{Sp})$ and write $\pi^* M = \pi_{-*} M$. A standard calculation gives¹

$$\pi^* R \cong \begin{cases} k[u_0, u_1, \dots], & p = 2, \quad |u_i| = 1, \\ \Lambda(e_0, e_1, \dots) \otimes k[v_0, v_1, \dots], & p > 2, \quad |e_i| = 1, |v_i| = 2. \end{cases}$$

Hence, with $d_n = 2n + 2$, the classes

$$a_n = \begin{cases} u_n^{2n+2}, & p = 2, \\ v_n^{n+1}, & p > 2, \end{cases} \quad a_n \in \pi^{d_n} R,$$

form a regular sequence. Represent a_n by $k \rightarrow k[d_n]$, and for $Y \in \mathcal{C}$ write $a_n^Y : Y \rightarrow Y[d_n]$ for its tensor action. Define

$$X^0 = k, \quad X^{n+1} = \text{fib}(X^n \xrightarrow{a_n^{X^n}} X^n[d_n]), \quad p_n : X^{n+1} \rightarrow X^n.$$

Put $w_n = d_n - 1$ and $s_n = \sum_{i < n} w_i$. From

$$\text{fib}(p_n) \simeq X^n[w_n]$$

one obtains inductively $X^n \in \mathcal{C}^{[-s_n, 0]}$. Hence, for fixed q , the map $\tau^{\geq -q} p_n$ is an equivalence once $w_n \geq q + 1$. Let X_q be the stable value of the eventually constant tower $\{\tau^{\geq -q} X^n\}_n$. The X_q are compatible under truncation and therefore define $X \in \widehat{\mathcal{C}}$. Limits in $\widehat{\mathcal{C}}$ are computed coordinatewise, so

$$X \simeq \varprojlim_n X^n.$$

Since $w_n \geq 1$, the zeroth coordinate never changes, and $\tau^{\geq 0} X \simeq k$; in particular $X \neq 0$.

Set $K^n = \text{RHom}_{\mathcal{C}}(k, X^n) \in \text{Mod}_R$ and $A = \pi^* R$. Then

$$K^0 \simeq R, \quad K^{n+1} \simeq \text{fib}(K^n \xrightarrow{a_n} K^n[d_n]).$$

Writing $A_n = A/(a_0, \dots, a_{n-1})$, the long exact sequence and regularity show inductively that $\pi^* K^n$ is a grading shift of A_n and that $\pi^* K^{n+1} \rightarrow \pi^* K^n$ is zero. Since $D(k)$ is semisimple, the underlying transition maps are themselves zero. The 1-shift formula for an inverse limit therefore gives

$$\varprojlim_n K^n \simeq 0$$

in $D(k)$, hence also in Mod_R because the forgetful functor preserves limits and detects zero objects.

Finally, X^n is bounded below, so the canonical functor $\mathcal{C} \rightarrow \widehat{\mathcal{C}}$ is fully faithful on k and X^n . Mapping out of k and using the preceding limit gives

$$\text{RHom}_{\widehat{\mathcal{C}}}(k, X) \simeq \varprojlim_n \text{RHom}_{\widehat{\mathcal{C}}}(k, X^n) \simeq \varprojlim_n K^n \simeq 0.$$

□

¹See E. Cline, B. Parshall, L. Scott, and W. van der Kallen, *Rational and generic cohomology*, Invent. Math. **39** (1977), 143–163, Theorem 4.1.

Weak generation in the heart and in the derived category

Proposition 2. *Let $\mathcal{A}$ be a Grothendieck abelian category and let $x \in \mathcal{A}$ satisfy $\mathrm{Hom}_{\mathcal{A}}(x, M) \neq 0$ for every $M \neq 0$. Then x generates $\mathcal{A}$ under kernels and cokernels of internal morphisms, extensions, and filtered colimits.*

Proof. Let $\mathcal{B}$ be a full replete subcategory with these closure properties and $x \in \mathcal{B}$. It contains 0 and internal images. Fix $K \subseteq x$ and choose by Zorn a maximal $L \subseteq K$ lying in $\mathcal{B}$; unions of chains remain in $\mathcal{B}$ because filtered colimits are exact. If $L \neq K$, take $0 \neq u : x \rightarrow K/L$ and postcompose with the monomorphism $K/L \hookrightarrow x/L$. Since $x, x/L \in \mathcal{B}$, its image I lies in $\mathcal{B}$; the monomorphism preserves the kernel, so $0 \neq I \subseteq K/L$. The inverse image of I in K is a strictly larger extension of I by L , contradicting maximality. Thus every subobject of x lies in $\mathcal{B}$.

Consequently, for every $f : x \rightarrow M$, $\mathrm{im}(f) = \mathrm{coker}(\ker(f) \hookrightarrow x)$ lies in $\mathcal{B}$. For arbitrary M , choose by Zorn a maximal subobject $N \subseteq M$ in $\mathcal{B}$. If $N \neq M$, a nonzero map $x \rightarrow M/N$ has nonzero image in $\mathcal{B}$, whose inverse image in M strictly enlarges N , again a contradiction. Hence $M \in \mathcal{B}$. $\square$

Theorem 3. *The analogous statement for the ordinary unbounded derived ∞ -category is false, even for x of finite projective dimension: there are $(\mathcal{A}, x)$ as above with $\mathrm{pd}_{\mathcal{A}}(x) \leq 2$ but $\mathrm{Loc}_{\mathcal{D}(\mathcal{A})}(x) \neq \mathcal{D}(\mathcal{A})$.*

Proof. Fix a field k and let $\mathcal{A}$ be the category of arbitrary-dimensional representations

$$M_1 \xrightarrow{a} M_2 \xrightarrow{b} M_1, \quad ba = 0,$$

the module category of the five-dimensional algebra $k(1 \xrightarrow{a} 2 \xrightarrow{b} 1)/(ba)$. Let S_1, S_2 be the simples and

$$P_1 = (k \xrightarrow{1} k \xrightarrow{0} k), \quad (P_2)_1 = k, \quad (P_2)_2 = kp \oplus kq, \quad a(1) = q, \quad b(p) = 1, \quad b(q) = 0.$$

Set $x = S_1 \oplus P_2$. Every nonzero module contains a simple submodule, and x surjects onto both simples; hence x maps nontrivially to every nonzero module. The projective resolution

$$0 \rightarrow P_1 \rightarrow P_2 \rightarrow P_1 \rightarrow S_1 \rightarrow 0$$

shows that $\mathrm{pd}_{\mathcal{A}}(x) \leq 2$.

Let $C^n = kc_n$ with zero differential and $E^n = kp_n \oplus kq_n$ with $d(p_n) = q_{n+1}$ and $d(q_n) = 0$. Define a complex Y by $Y_1 = C$, $Y_2 = E$, and

$$a(c_n) = q_n, \quad b(p_n) = c_n, \quad b(q_n) = 0.$$

The complex E is acyclic, while $H^n(Y) \simeq S_1$ for every n , so $Y \neq 0$. Since P_2 represents evaluation at vertex 2, $\mathrm{RHom}_{\mathcal{A}}(P_2, Y) \simeq E \simeq 0$. Applying $\mathrm{Hom}_{\mathcal{A}}(-, Y)$ to the displayed resolution computes $\mathrm{RHom}_{\mathcal{A}}(S_1, Y)$ as the finite totalization of the degreewise exact sequence

$$0 \rightarrow C \xrightarrow{a} E \xrightarrow{b} C \rightarrow 0,$$

so it is acyclic. Thus $\mathrm{RHom}_{\mathcal{A}}(x, Y) \simeq 0$ although $Y \neq 0$. $\square$

Unseparated variant. For a singular affine noetherian scheme Z , Lurie's unseparated derived category is isomorphic to $\mathrm{IndCoh}(Z)$, where $\mathcal{O}_Z^{\frac{1}{2}} \neq 0$ is clear; see *Spectral Algebraic Geometry*, Appendix C.5.